MVM — Mutation Validation Module

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Compatibility:
INTEGROS® · SESCS · MLM · CGRM · RTI · MTVF · EVIP ·
ORGS (OGS-VFM Level 0)

Authority: OOF™ Origin Open Foundation™
Protection: MIP™ — Methodological Intellectual Property
Canonical Language: English (UCL™)

Canonical Definition System

Mutation Validation defines the **non-bypassable conditions under which any system change (mutation) is evaluated, validated, and classified before it is allowed to affect system behavior.**

A mutation is any change to:

- system logic
- model behavior
- data structure
- execution pathways

No mutation is valid if it cannot be verified, classified, and approved according to defined integrity conditions.

A. Module Abstract

The Mutation Validation Module ensures that system evolution does not compromise integrity.

The objective is to ensure that:

- changes are not only recorded (MLM)
- but also validated before impact
- system evolution is controlled
- unsafe mutations are blocked

This module transforms mutation from an uncontrolled event into a governed process.

B. Module Purpose

The purpose of this module is to prevent uncontrolled system evolution.

Without mutation validation:

- systems change unpredictably
- harmful changes propagate
- integrity may be broken silently
- trust collapses

This module ensures that every change is evaluated before it becomes real.

C. Core Principle

A system may change only if the change is validated.

D. Structural Requirements

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1. Mutation Identification

Every mutation MUST be identifiable.

The system MUST detect:

- when a change occurs
- what component is affected
- what type of mutation is executed

Undetected mutation is invalid.

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2. Mutation Classification

Each mutation MUST be classified.

Minimum classification:

- low impact
- high impact
- critical mutation

Classification MUST determine validation requirements.

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3. Validation Requirement

Every mutation MUST be validated before activation.

Validation may include:

- rule-based validation
- integrity checks
- governance approval

- system constraints

Unvalidated mutation MUST NOT affect system behavior.

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4. Validation Outcome

Every mutation MUST produce a result:

- approved
- rejected
- conditional

Rejected mutation MUST NOT execute.
Conditional mutation MUST operate under defined limits.

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5. Integrity Impact Control

The system MUST evaluate:

- whether the mutation affects integrity
- whether traceability is preserved
- whether system consistency is maintained

Mutations that reduce integrity MUST be blocked.

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6. Pre-Execution Enforcement

Validation MUST occur before the mutation becomes active.

The system MUST prevent:

- post-change validation as primary control
- delayed validation
- silent mutation activation

E. Minimum Implementation Framework (MIF)

Step 1 — Detect Mutation

The system MUST detect all changes.

Minimum condition:

- mutation trigger is captured

Step 2 — Link to Mutation Log (MLM)

The system MUST connect mutation to recorded event.

Minimum condition:

- mutation is logged with identifier and timestamp

Step 3 — Classify Mutation

The system MUST determine mutation impact.

Minimum condition:

- mutation is categorized

Step 4 — Validate Mutation

The system MUST evaluate mutation before activation.

Minimum condition:

- validation decision exists

Step 5 — Enforce Outcome

The system MUST:

- allow approved mutations
- block rejected mutations
- control conditional mutations

F. Validation Logic

A mutation is valid only if:

- it is detected
- it is classified
- it is validated
- it does not break integrity
- it is linked to traceable record

If any condition fails, the mutation is invalid.

G. Prohibited Conditions

A system is non-compliant if:

- mutations occur without validation
- classification is missing
- rejected mutations still execute
- validation occurs after execution
- integrity impact is ignored

No system may claim integrity if it allows unvalidated change.

H. Operational Output

A system implementing this module produces:

- controlled system evolution
- validated change processes

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Canonical source: https://originopenfoundation.org/content/s/sescs-mvm.html

- integrity-preserving updates
- predictable system behavior

I. System Condition

The system is:

Valid if all mutations are validated before activation
Invalid if any mutation affects behavior without validation

There is no partial mutation control.

J. Integration Logic

This module operates together with:

- INTEGROS® — ensures integrity conditions
- MLM — records all mutations
- SESCS — governs self-evolving systems
- CGRM — defines authority
- RTI — ensures responsibility for changes
- MTVF — validates truth of mutation
- EVIP — controls permission

MLM records change. MVM validates change.

K. Use Case 1 — AI Model Drift

Scenario

AI model updates its internal logic.

Without MVM

- change occurs silently
- behavior shifts
- no validation
- unpredictable outcomes

With MVM

- mutation detected
- classified as high impact
- validated before activation

Result

- system evolution controlled
- behavior predictable
- integrity preserved

L. Use Case 2 — Malicious Code Injection

Scenario

External or internal change modifies system logic.

Without MVM

- change executes immediately
- integrity compromised
- no detection

With MVM

- mutation detected
- fails validation
- execution blocked

Result

- attack prevented
- system remains stable
- integrity maintained

M. Structural Principle

Change is not risk.
Unvalidated change is risk.

Canonical Closing Statement

If a change cannot be validated, it cannot be allowed.

Mutation Validation Module establishes a non-bypassable condition:

every mutation must be validated before it becomes real.

Related Documents

→ SESCS Standard

→ SESCS — Master Index

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Canonical Source

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